Hand-set parameters: v18 and v20

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Every parameter below is fixed by hand. "Measured" means project results support the value; "Design" means it rests on reasoning or physics and was not tested separately.

Validation and training data

Parameter	Value	Model	Justification	Basis
Local train/test split	First 80% / last 20%, by time	Both	The test period follows training, as on Kaggle; random shuffling would leak overlapping weeks	Design
Gap before local test	7 cases	Both	Each case spans 7 days, so the gap prevents overlap with training	Design
CV folds	5 contiguous folds	Both	Provides out-of-fold predictions for stacking with an acceptable training time	Design
Gap around CV folds	7 cases in v18; none in v20	Both	Removes overlapping weeks in v18. v20 keeps AutoGluon's grouped folds, so its out-of-fold scores can be slightly optimistic	Design
Recent-case weight	2.0 for cases from 2021 onward, Kaggle runs only	Both	Market conditions changed around that period; earlier years keep about 60% of the total weight. On v12 the public score rose from 0.98769 to 0.98811, comparable to noise	Measured, weak

Feature constants

Both models share these values.

Parameter	Value	Justification	Basis
Volume-flow conversion	1 m^3/s for 1 h = 0.0036 Mm^3	Unit conversion	Physics
Plant capacity formula constants	2.725 and efficiency 0.88	2.725 converts head and flow to MW per Mm^3/h; 0.88 is an assumed typical overall efficiency for a fast capacity estimate	Physics, assumed efficiency
Generator efficiency	0.98	Single value near the top of the 97-99% range in the topology file	Design
Minimum head	5 m in capacity, 1 m for units	Prevents division by zero or negative head	Numerical guard
Log floor	Prices and water values clipped at 1 before the log	Makes log ratios defined for near-zero or negative values	Numerical guard
Normalization offsets	+1 on price sd, +5 on water value	Prevents very large ratios when the denominator is close to zero	Numerical guard
Upstream lags	1 h and 3 h	Gives the plant recent upstream activity; values not tuned separately	Design
Surrogate budgets	4: minimum to avoid spill, water value at start, at end, maximum usable	Covers the range from forced release to all usable water. The surrogate raised local test AUC from 0.9438 to 0.9570 in v9	Measured
Water value source	Vinje uses Totak; Haukeli uses Langeidvatn	Vatjern's volume is constant in the data, and Totak supplies Vaamarvatn through a tunnel	Data and topology

Parameter	Value	Justification	Basis
Viterbi switching cost	Half of start + stop cost per state change	Keeps the same total over a full on/off cycle	Design
LP tunnel limit	±500 m³/s for reversible tunnels, 0 to 500 otherwise	The topology gives no tunnel limit, so this is set high enough not to bind	Design
LP slack penalty	10000	Keeps minimum constraints soft but much costlier than any market value, so the LP stays feasible	Design
LP cost on curved rivers	2	Linear stand-in for the topology's flow cost curves	Design

Adding the unit physics and start-cost features raised local test AUC from 0.9807 to 0.9823 in v14. Adding the LP took it to 0.9825, and Haukeli rose from 0.8983 to 0.9227.

Tree branches

These are conventional starting values, kept fixed. Dedicated tuning on an earlier XGBoost version did not help: random search gave 0.9261 and a second search 0.9393, against 0.9423 with the fixed settings.

Parameter	Value	Model	Justification	Basis
Trees, learning rate	200, 0.05	XGBoost	Small steps with a moderate number of trees; compatible with the time limit	Measured via earlier tuning
Maximum depth	6	XGBoost	Allows interactions without very specific rules	Measured via earlier tuning
Row and column sampling	0.8, 0.8	XGBoost	Reduces overfitting, relevant because many features are correlated	Measured via earlier tuning

Parameter	Value	Model	Justification	Basis
Seeds	3	XGBoost	Averaging reduces run-to-run noise, measured at about 0.0015 on v4-C	Measured
LightGBM	300 rounds, learning rate 0.05, 63 leaves, feature fraction 0.5, bagging 0.8	AutoGluon	Same learning rate as XGBoost, with stronger column sampling to differ from it	Design
CatBoost	300 iterations, learning rate 0.05, depth 6	AutoGluon	Same scale as XGBoost, with a different tree structure	Design
ExtraTrees	100 trees, depth 16, min 20 samples per leaf, max features 0.3	AutoGluon	Randomized splits add diversity; depth and leaf size limit overfitting and memory	Design
CPUs	4	AutoGluon	Project limit	Project rule
Stack levels	0	AutoGluon	A weighted ensemble of base models keeps training time and overfitting risk low	Design

TCN

Both models use the same network and settings.

Parameter	Value	Justification	Basis
Kernel size, dilations	3; 1, 2, 4, 8, 24, 48, 72	Short dilations capture nearby hours, and 24-72 h capture daily patterns. The receptive field of 319 h covers the whole week from every hour	Design

Parameter	Value	Justification	Basis
Non-causal convolutions	Yes	The week's inputs are known before the schedule is made	Design
Channels	128	Compresses up to 547 input channels; a larger 192-channel variant was prepared but not adopted	Design
GroupNorm groups	8	Works with small batches	Design
Dropout	0.2, also on whole input channels	Limits reliance on single channels, since many are correlated	Design
Learning rate, weight decay	1e-3, 1e-2	Standard AdamW values, with a one-cycle schedule	Design
Epochs, batch size	10, 32	Fits the time budget. Test AUC was logged only as a diagnostic and not used to pick the epoch	Design
Seeds	3	Averaging logits reduces noise: single seeds reached 0.9706-0.9722, the average 0.9723	Measured
Price scale copies	×0.5, ×1, ×2, ×4	Makes the model less sensitive to price level; the local test shows why this matters, with 94.1% of its cases outside the training price range	Design, supported by data
Input clipping	±8 after z-score	Limits extreme values during price spikes	Numerical guard
Constant-channel threshold	sd ⩽ 1e-4 removed	Removes channels with no information and rounding noise	Numerical guard

Final combination

These values were selected on the local test set, so their local scores are slightly optimistic.

Parameter	Value	Model	Justification	Basis
Meta-model regularization	C = 1e-4	v18	CV / forward AUC: C = 1e-2 gave 0.9866 / 0.9867, C = 1e-4 gave 0.9862 / 0.9869, C = 1e-6 gave 0.9837 / 0.9843. It has the best forward score and limits per-unit corrections	Measured
Meta-model inputs	Logits, not standardized	v18	Standardized inputs were clearly worse, with CV AUC of 0.944 or lower	Measured
Meta-model folds	5 folds, 7-case gap	v18	Cases are scored by a model that did not see them, and overlapping weeks are removed	Design
TCN weight	0.2	v20	Best value on a 0.1-step grid: 0.9848, against 0.9830 with no TCN and 0.9844 at 0.1 and 0.3	Measured